

TESTING – TEST PLANS AND TEST CASES

Aim:

To create Test Plans and Test Cases for the Online Appointment Booking System for Clinics, and to write two test cases for at least five user stories showcasing the happy path and error scenarios in the Azure DevOps platform.

Procedure:

1. Understand Core Features of the Application

- Patient Registration & Secure Login
- Appointment Booking by Doctor Specialization
- Doctor Schedule Management
- Appointment Notifications (Confirmations, Reminders, Cancellations)
- Secure Data Handling and System Monitoring (Admin)

2. Define User Interactions

- Each test case simulates real user behaviour (e.g., registering a patient, booking an appointment, updating a doctor schedule).

3. Design Happy Path Test Cases

- Focused on validating that all features function as expected under normal conditions.
- Example: Patient registers successfully, books an appointment, and receives a confirmation notification.

4. Design Error Path Test Cases

- Simulate negative or unexpected scenarios to test system robustness and error handling.
- Example: Login fails with an incorrect password, booking fails when no time slot is selected, unauthorized admin access is blocked.

5. Break Down Steps and Expected Results

- Each test case contains step-by-step actions and a corresponding expected outcome.
- Ensures clarity for both manual testers and automation scripts.

6. Use Clear Naming and IDs

- Test cases are named clearly (e.g., TC01 – Successful Registration, TC12 – Unauthorized Access).
- Helps in quick identification and linking to user stories or features.

7. Separate Test Suites

- Grouped test cases based on functionality (e.g., Registration, Booking, Notifications, Admin Monitoring).
- Improves organisation and test execution flow in Azure DevOps.

8. Prioritize and Review

- Critical user actions are marked high-priority.
- Reviewed for completeness and traceability against feature requirements.

New Test Plan

The screenshot shows the 'New Test Plan' form in Azure DevOps. The left sidebar contains a navigation menu with options: Overview, Boards, Repos, Pipelines, Test Plans (selected), Test plans, Progress report, Parameters, Configurations, Runs, Exploratory sessions, Artifacts, and Project settings. The main form area has the following fields:

- Name ***: A text input field with the placeholder 'Enter a plan name'.
- Area Path ***: A dropdown menu with 'PROJECT_01' selected.
- Iteration ***: A dropdown menu with 'PROJECT_01/Iteration 1' selected.

At the bottom right of the form are two buttons: 'Create' and 'Cancel'.

Test Suite

The screenshot shows the 'Test Suite' view in Azure DevOps. The left sidebar is the same as in the previous screenshot. The main content area is titled 'Appointment_Booking_System' and shows a list of test suites under the 'Test Suites' section. The list includes:

- User Management
- Appointment Management
- Doctor Management
- Clinic Management
- Notification & Reminders
- Payment Management
- Reports & Analytics
- User Login (1)

Below the list, there is a table titled 'Test Cases (1 item)'. The table has columns: Title, Order, Test Case Id, Assigned To, and State. The table contains one row:

Title	Order	Test Case Id	Assigned To	State
Successful login with valid credentials	1	24	Sunil Varma	Design

At the top right of the table, there is a 'New Test Case' button. The right sidebar shows the 'User Login (ID: 24)' section with tabs for 'Define', 'Execute', and 'Chart'.

Test Plan Setup in Azure DevOps:

• New Test Plan

In Azure DevOps, navigate to Test Plans from the left navigation menu. Click on New Test Plan. Provide the following details:

- Name: Appointment_Booking_System Test Plan
- Area Path: Appointment_Booking_System
- Iteration: Sprint 1 (April 26 – May 10, 2026)

Click Create to set up the test plan.

• Test Suite

Under the test plan, create separate Test Suites for each user story module. Right-click the test plan and select Add Static Test Suite. Name each suite according to the feature being tested.

User Stories:

TS01 – User Registration & Login (US-01): As a patient, I want to register and log in securely so that I can access my appointment dashboard.

TS02 – Appointment Booking by Specialization (US-02): As a patient, I want to book an appointment by selecting a specialization and available doctor so that I can get the right medical attention.

TS03 – Doctor Schedule Management (US-03): As a doctor, I want to manage my schedule and view upcoming appointments so that I can organize my clinic hours efficiently.

TS04 – Appointment Notifications (US-04): As a patient, I want to receive notifications for my appointment confirmation, reminders, and cancellations so that I stay informed.

TS05 – Secure Data Handling & System Monitoring (US-05): As an admin, I want the system to securely store patient data and monitor system performance so that data privacy and reliability are ensured.

Test Cases:

Test Suite: TS01 – User Registration & Login (US-01)

TC01 – Successful Patient Registration

Type: Happy Path

Action:

- Navigate to the clinic booking portal registration page.

- Enter valid full name, email address, phone number, and a strong password.
- Click the 'Register' button.

Expected Results:

- Registration form is displayed correctly.
- All fields accept valid input without errors.
- Account is created successfully and user is redirected to the login page with a confirmation message.

TC02 – Successful Patient Login

Type: Happy Path

Action:

- Navigate to the login page.
- Enter valid registered email and correct password.
- Click the 'Login' button.

Expected Results:

- Login form is displayed with email and password fields.
- Credentials are accepted.
- User is authenticated and redirected to the appointment dashboard.

TC03 – Registration with Already Registered Email

Type: Error Path

Action:

- Navigate to the registration page.
- Enter an already registered email address along with other valid details.
- Click the 'Register' button.

Expected Results:

- Form fields accept input.
- System returns error: 'Email address is already registered. Please log in.'
- Registration is not completed.

TC04 – Login with Incorrect Password

Type: Error Path

Action:

- Navigate to the login page.
- Enter a valid registered email with an incorrect password.
- Click the 'Login' button.

Expected Results:

- Input fields accept data.
- System returns error: 'Invalid email or password. Please try again.'
- User is not logged in; login page remains displayed.

Test Suite: TS02 – Appointment Booking by Specialization (US-02)

TC05 – Book Appointment with Valid Specialization and Doctor

Type: Happy Path

Action:

- Log in as a registered patient.
- Navigate to 'Book Appointment'.
- Select a specialization (e.g., Cardiology), choose an available doctor, select a date and time slot.
- Click 'Confirm Appointment'.

Expected Results:

- Specialization and doctor lists are populated correctly.
- Available slots are shown for the selected doctor.
- Appointment is booked and a confirmation notification is displayed with appointment ID.

TC06 – Book Appointment Without Selecting a Time Slot

Type: Error Path

Action:

- Log in as a registered patient.
- Navigate to 'Book Appointment'.
- Select a specialization and doctor but skip selecting a time slot.
- Click 'Confirm Appointment'.

Expected Results:

- System validates the form.
- Error message displayed: 'Please select an available time slot to proceed.'
- Appointment is not booked.

Test Suite: TS03 – Doctor Schedule Management (US-03)

TC07 – Doctor Views and Updates Schedule

Type: Happy Path

Action:

- Log in as a doctor.
- Navigate to 'My Schedule'.
- Set available days and hours for the upcoming week.
- Click 'Save Schedule'.

Expected Results:

- Schedule management page loads successfully.
- Doctor can select days and set time slots.
- Schedule is saved and a success message is shown: 'Schedule updated successfully.'

TC08 – Doctor Saves Schedule with Overlapping Slots

Type: Error Path

Action:

- Log in as a doctor.
- Navigate to 'My Schedule'.
- Attempt to add two appointments at the same date and time.
- Click 'Save Schedule'.

Expected Results:

- System detects the time conflict.
- Error shown: 'Time slot conflict detected. Please adjust your schedule.'
- Conflicting schedule is not saved.

Test Suite: TS04 – Appointment Notifications (US-04)**TC09 – Patient Receives Confirmation Notification After Booking**

Type: Happy Path

Action:

- Log in as a patient and complete a valid appointment booking.
- Check registered email inbox and the in-app notification panel.

Expected Results:

- Appointment is created successfully.
- Patient receives an email and in-app notification containing appointment details (doctor name, date, time, location).

TC10 – Notification Not Delivered Due to Invalid Email

Type: Error Path

Action:

- Register a patient account with an improperly formatted email address.
- Complete an appointment booking.
- Check the notification delivery status in the admin panel.

Expected Results:

- Booking is completed in the system.
- Email notification fails to deliver.
- Admin panel shows delivery failure with error: 'Invalid recipient email address.'

Test Suite: TS05 – Secure Data Handling & System Monitoring (US-05)**TC11 – Admin Views System Monitoring Dashboard**

Type: Happy Path

Action:

- Log in as an admin user.
- Navigate to the 'System Monitoring' section.
- Observe the performance metrics panel (CPU usage, active users, response times).

Expected Results:

- Admin dashboard loads with real-time system metrics.
- All health indicators are within acceptable thresholds.
- Metrics refresh automatically without page reload.

TC12 – Unauthorized Access to Patient Data

Type: Error Path

Action:

- Log in as a regular patient user.
- Attempt to access the admin data management panel by directly navigating to the admin URL.
- Observe the system response.

Expected Results:

- Access is denied.
- System returns error: 'Access Forbidden – You do not have permission to view this resource.'
- Unauthorized access attempt is logged in the system audit trail.

Azure DevOps

dev.azure.com/pradhikshyaanand / Online_Appointment_Booking_System / Test Plans / Online Appointment Booking System

Search

5

Online_Appointment_Booking_System

Overview

Boards

Repos

Pipelines

Test Plans

Test plans

Progress report

Parameters

Configurations

Runs

Exploratory sessions

Artifacts

Project settings

Online Appointment Booking System

May 16 - May 30

Past

15% run exit

1% passed [View report](#)

Test Suites

Filter suites by name

Online Appointment Booking System

User Management

Appointment Management

Doctor Management

Clinic Management

Notification & Reminders

Payment Management

Reports & Analytics

User Login (1)

User Login (ID: 24)

Define

Execute

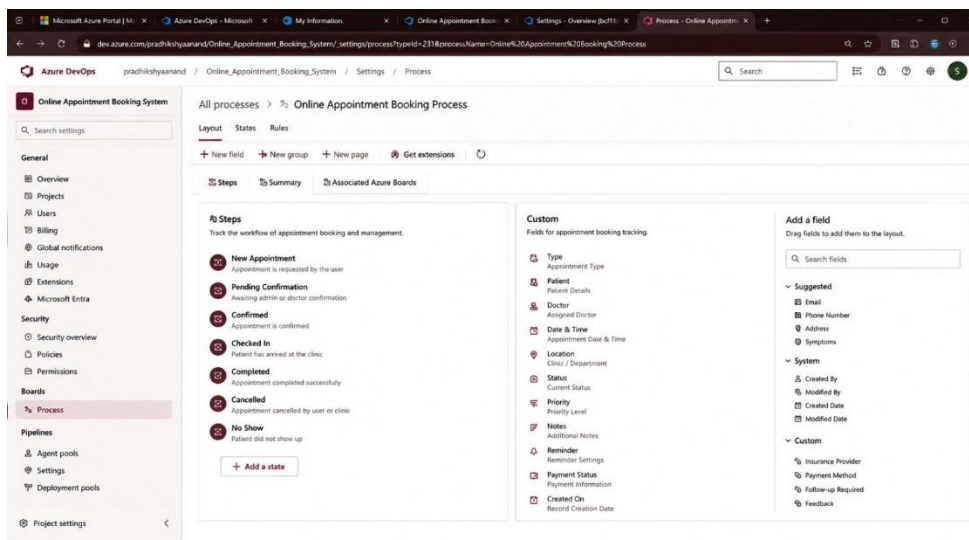
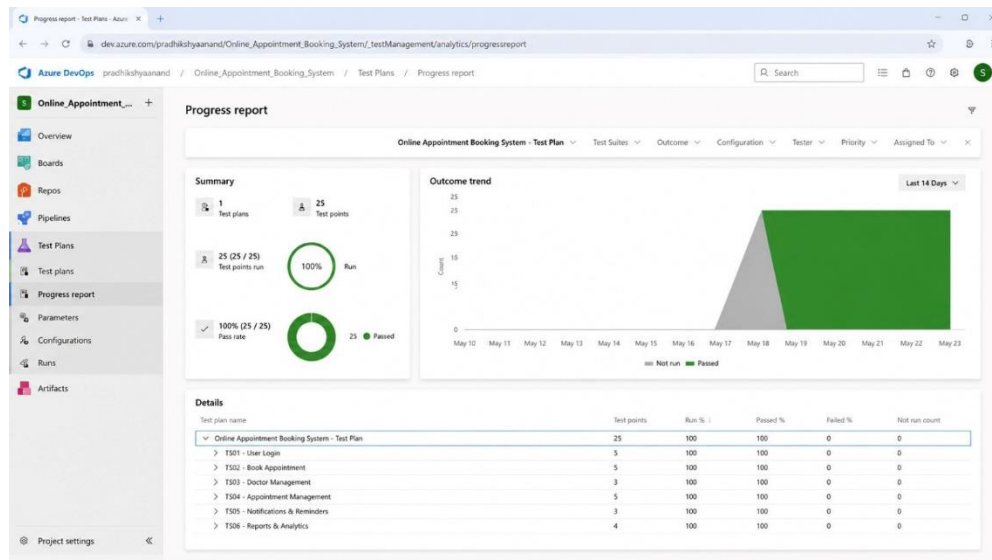
Chart

Test Cases (1 item)

New Test Case

☐ Title	Order	Test Case Id	Assigned To	State
☐ Successful login with valid credentials	1	24	Saranya	Design

Progress Report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path